Fetal Health Prediction

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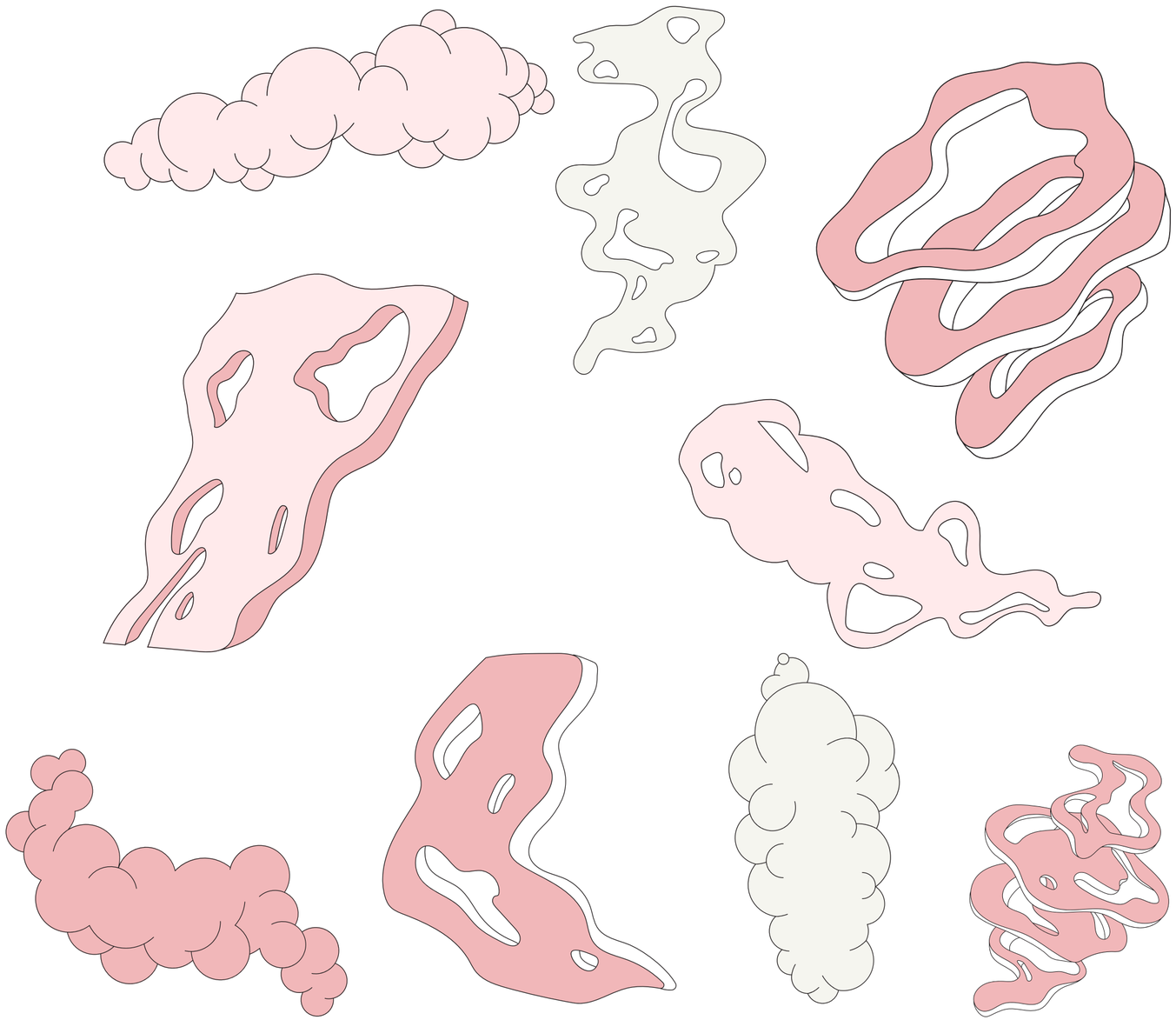
Overview & Business problem

Understanding Infant Mortality and Cardiotocography (CTG)



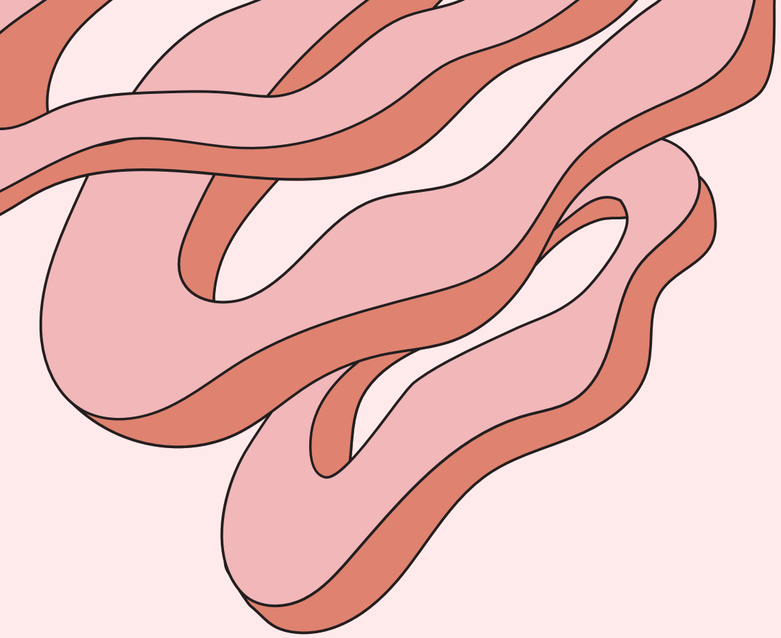
Infant death

The infant mortality rate is 24 per 1,000

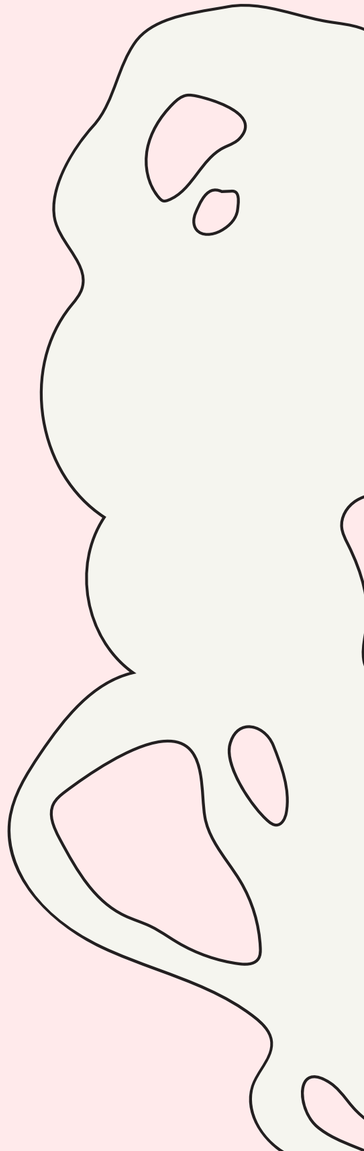
If 100 people give birth, between 2 and 3 die

The health of pregnant women and their fetuses is interrelated

An unhealthy fetus is at risk of death during childbirth



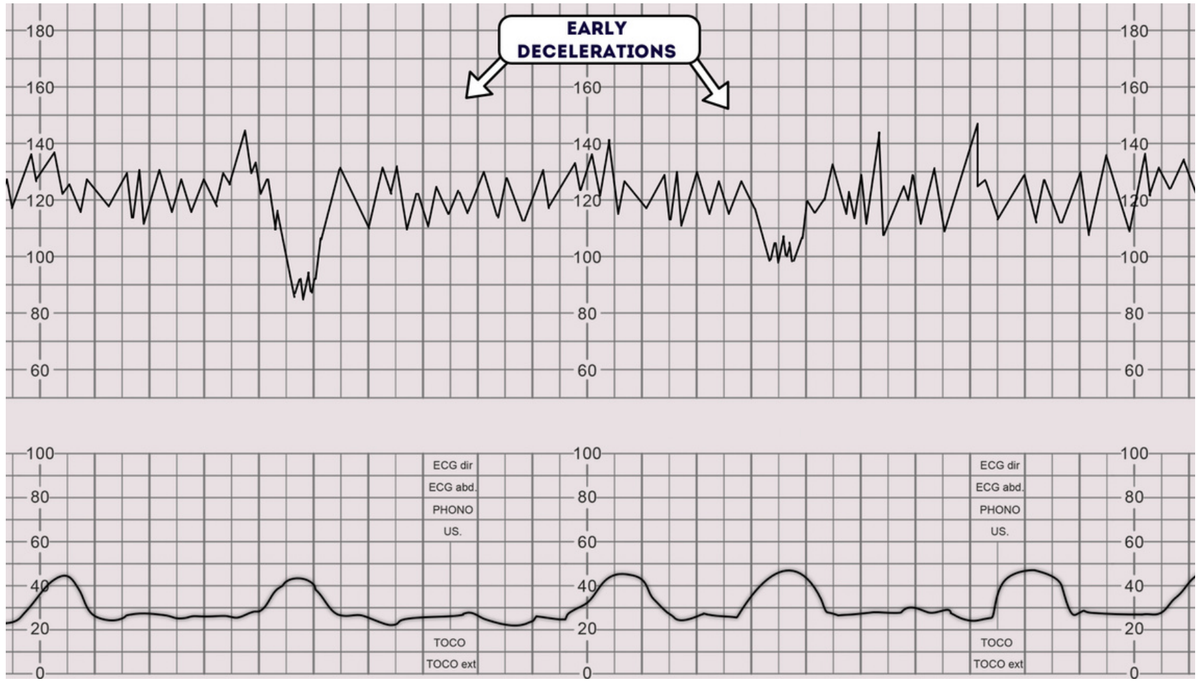
!! **Pregnant women do not know health condition of the unborn fetus**



Metode Cardiotocography (CTG)

The Cardiotocography (CTG) method is the process of examining the health of the fetus in pregnant women.

To monitor activity, fetal heart rate, uterine contractions and several elements needed when the fetus is in the womb.

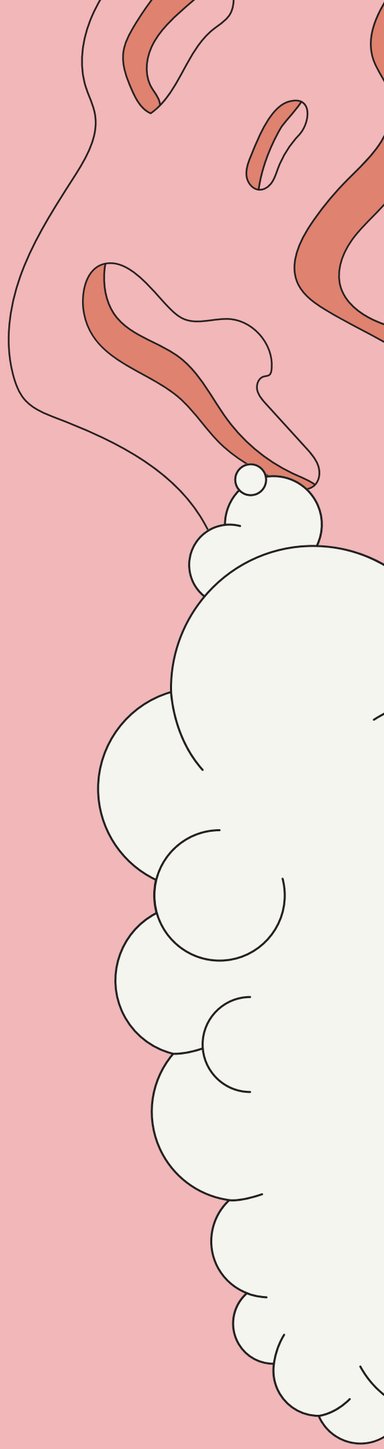


Cardiotocograms

- Difficult to interpret
- Interpreted Fetal Health Specialist

How does a data scientist play a role in reducing infant mortality?

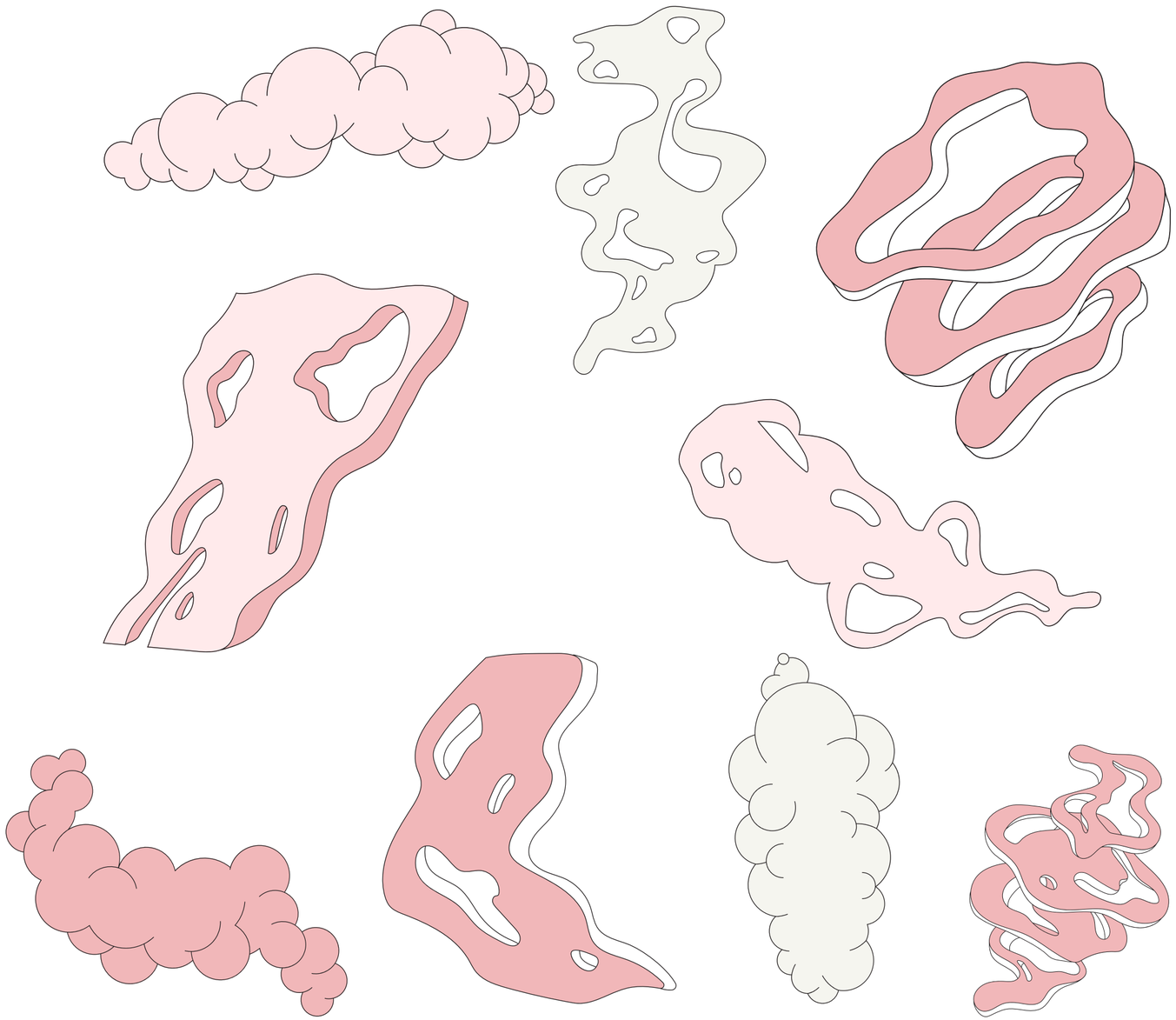
Building machine learning predictions about the classification of fetal health status using cardiotocography data



2

Data understanding & Exploratory Data Analysis

Understand and take insight from data



DATASET

Contains
2126 rows & 22 Columns

link dataset:
https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification

The 10 columns describe the shape of the histogram

11 columns explain the basic things in evaluating CTG results

Fetal_health column as target
1 (Normal), 2 (Suspect) and 3 (Pathological)

Basic CTG Evaluation Columns

- baseline_value: FHR baseline (beats per minute)
- accelerations: Number of accelerations per second
- fetal_movement: Number of fetal movements per second
- uterine_contractions: Number of uterine contractions per second
- light_decelerations: Number of light decelerations per second
- severe_decelerations: Number of severe decelerations per second
- prolongued_decelerations: Number of prolonged decelerations per second
- abnormal_short_term_variability: Percentage of time with abnormal short term variability
- mean_value_of_short_term_variability: Mean value of short term variability
- percentage_of_time_with_abnormal_long_term_variability: Percentage of time with abnormal long term variability
- mean_value_of_long_term_variability: Mean value of long term variability

Histogram Shape Columns

- histogram_width: Width of FHR histogram
- histogram_min: Minimum (low frequency) of FHR histogram
- histogram_max: Maximum (high frequency) of FHR histogram
- histogram_number_of_peaks: Number of histogram peaks
- histogram_number_of_zeroes: Number of histogram zeros
- histogram_mode: Histogram mode
- histogram_mean: Histogram mean
- histogram_median: Histogram median
- histogram_variance: Histogram variance
- histogram_tendency: Histogram tendency

General Information

#	Column	Non-Null Count	Dtype
0	baseline value	2126 non-null	float64
1	accelerations	2126 non-null	float64
2	fetal_movement	2126 non-null	float64
3	uterine_contractions	2126 non-null	float64
4	light_decelerations	2126 non-null	float64
5	severe_decelerations	2126 non-null	float64
6	prolongued_decelerations	2126 non-null	float64
7	abnormal_short_term_variability	2126 non-null	float64
8	mean_value_of_short_term_variability	2126 non-null	float64
9	percentage_of_time_with_abnormal_long_term_variability	2126 non-null	float64
10	mean_value_of_long_term_variability	2126 non-null	float64
11	histogram_width	2126 non-null	float64
12	histogram_min	2126 non-null	float64
13	histogram_max	2126 non-null	float64
14	histogram_number_of_peaks	2126 non-null	float64
15	histogram_number_of_zeroes	2126 non-null	float64
16	histogram_mode	2126 non-null	float64
17	histogram_mean	2126 non-null	float64
18	histogram_median	2126 non-null	float64
19	histogram_variance	2126 non-null	float64
20	histogram_tendency	2126 non-null	float64
21	fetal_health	2126 non-null	float64

Missing Values **0**

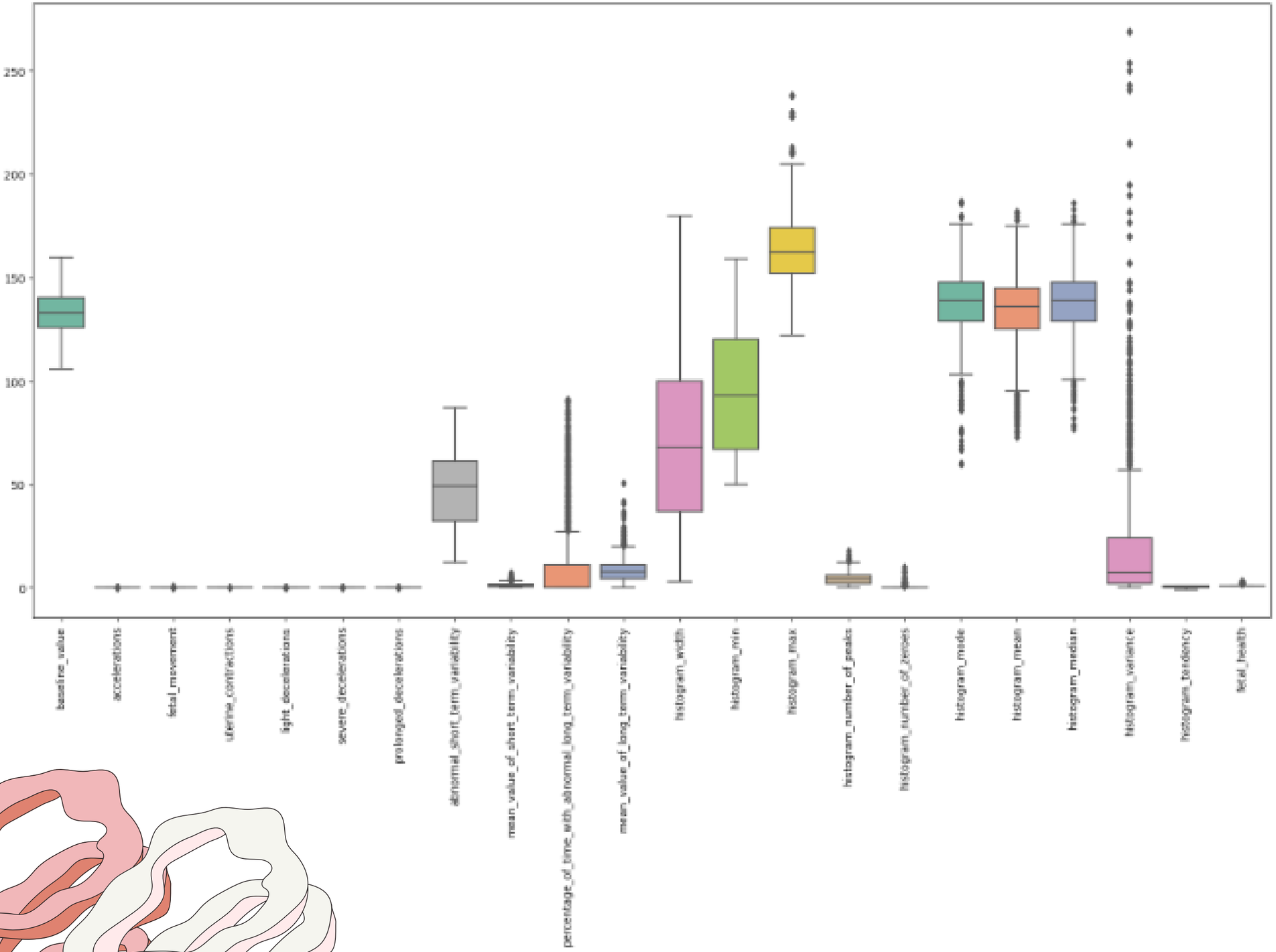
Duplicate **13**

All Numerical data type

Statistical Summary

	count	mean	std	min	25%	50%	75%	max
baseline_value	2113.0	133.304780	9.837451	106.0	126.000	133.000	140.000	160.000
accelerations	2113.0	0.003188	0.003871	0.0	0.000	0.002	0.006	0.019
fetal_movement	2113.0	0.009517	0.046804	0.0	0.000	0.000	0.003	0.481
uterine_contractions	2113.0	0.004387	0.002941	0.0	0.002	0.005	0.007	0.015
light_decelerations	2113.0	0.001901	0.002966	0.0	0.000	0.000	0.003	0.015
severe_decelerations	2113.0	0.000003	0.000057	0.0	0.000	0.000	0.000	0.001
prolonged_decelerations	2113.0	0.000159	0.000592	0.0	0.000	0.000	0.000	0.005
abnormal_short_term_variability	2113.0	46.993848	17.177782	12.0	32.000	49.000	61.000	87.000
mean_value_of_short_term_variability	2113.0	1.335021	0.884368	0.2	0.700	1.200	1.700	7.000
percentage_of_time_with_abnormal_long_term_variability	2113.0	9.795078	18.337073	0.0	0.000	0.000	11.000	91.000
mean_value_of_long_term_variability	2113.0	8.166635	5.632912	0.0	4.600	7.400	10.800	50.700
histogram_width	2113.0	70.535258	39.007706	3.0	37.000	68.000	100.000	180.000
histogram_min	2113.0	93.564600	29.562269	50.0	67.000	93.000	120.000	159.000
histogram_max	2113.0	164.099858	17.945175	122.0	152.000	162.000	174.000	238.000
histogram_number_of_peaks	2113.0	4.077142	2.951664	0.0	2.000	4.000	6.000	18.000
histogram_number_of_zeroes	2113.0	0.325603	0.707771	0.0	0.000	0.000	0.000	10.000
histogram_mode	2113.0	137.454330	16.402026	60.0	129.000	139.000	148.000	187.000
histogram_mean	2113.0	134.599621	15.610422	73.0	125.000	136.000	145.000	182.000
histogram_median	2113.0	138.089446	14.478957	77.0	129.000	139.000	148.000	186.000
histogram_variance	2113.0	18.907241	29.038766	0.0	2.000	7.000	24.000	269.000
histogram_tendency	2113.0	0.318504	0.611075	-1.0	0.000	0.000	1.000	1.000
fetal_health	2113.0	1.303833	0.614279	1.0	1.000	1.000	1.000	3.000

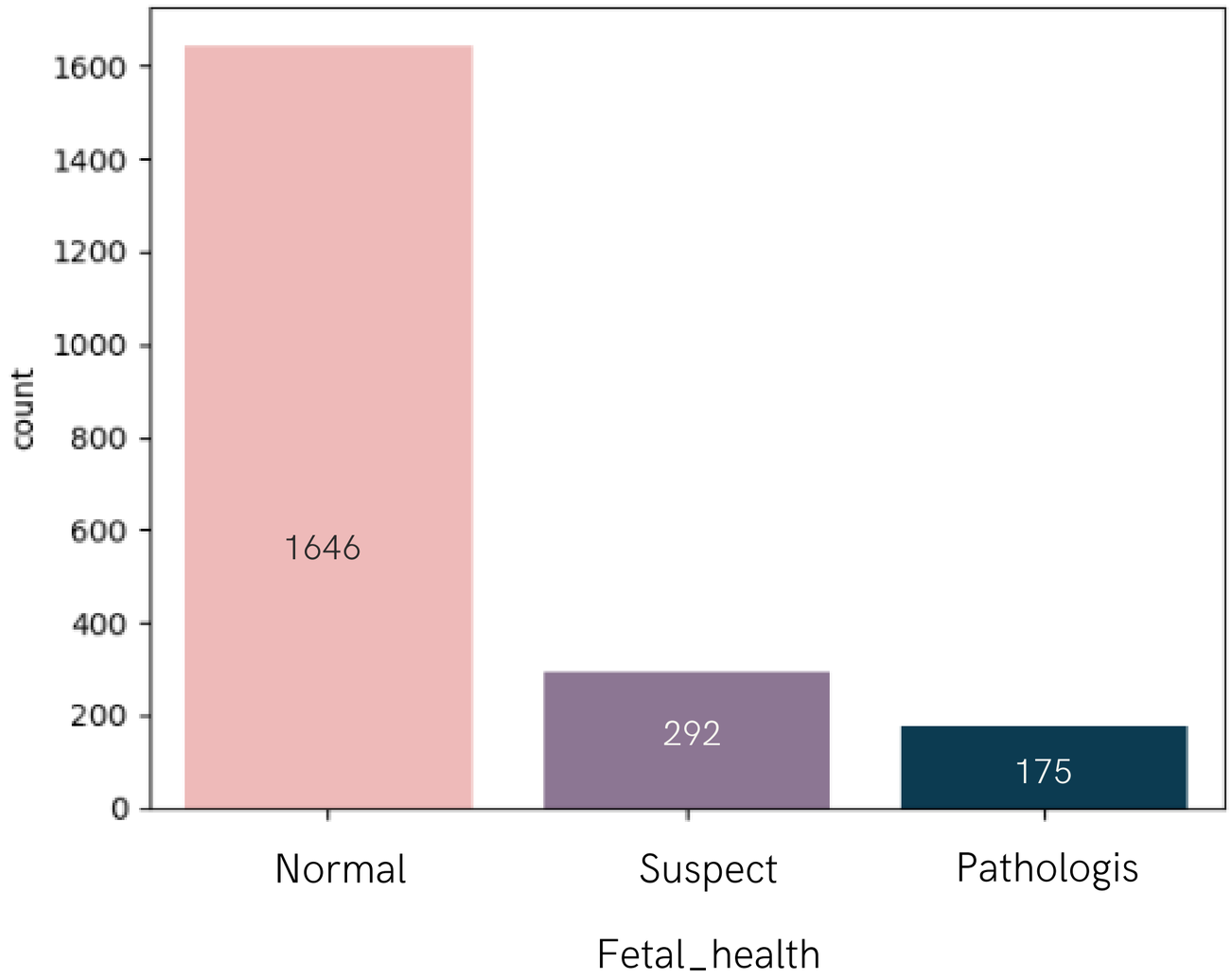
- Min-max values are mostly making sense
- Just the baseline_value column is somewhat symmetrical, mean ~ median.



Univariate Analysis

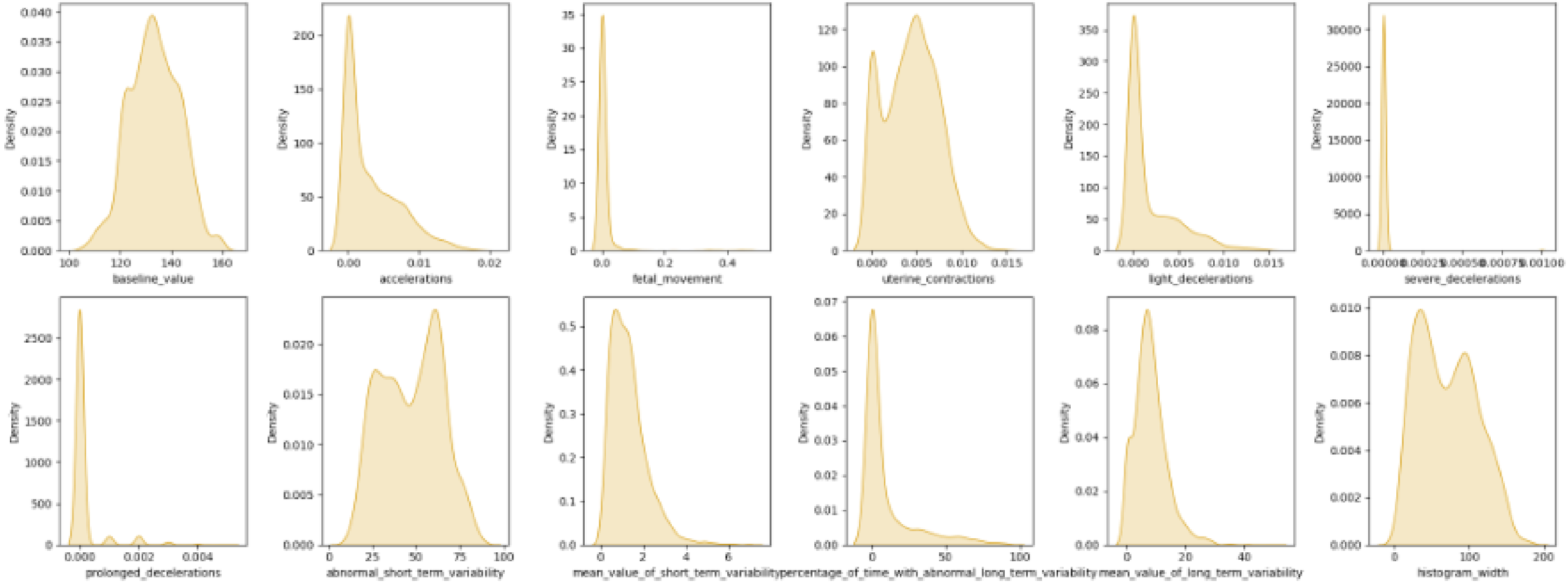
- There are many columns with outliers
- It's all about the fetus and the experts marking its classification. So, no need to drop
- All the features are in different ranges

Target Column



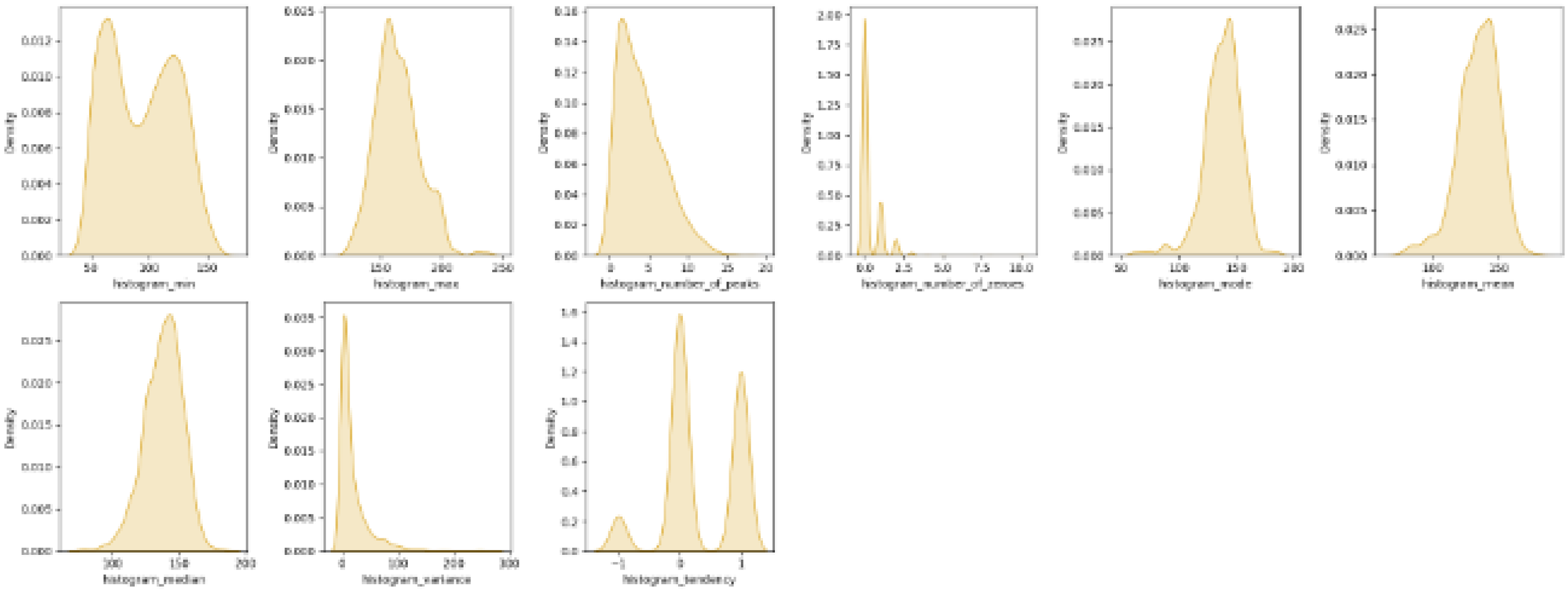
- The majority of fetuses in this dataset are normal
- Normal fetus as much as 1646
- 292 suspected fetuses
- Pathological fetuses as many as 175

KDE Plot



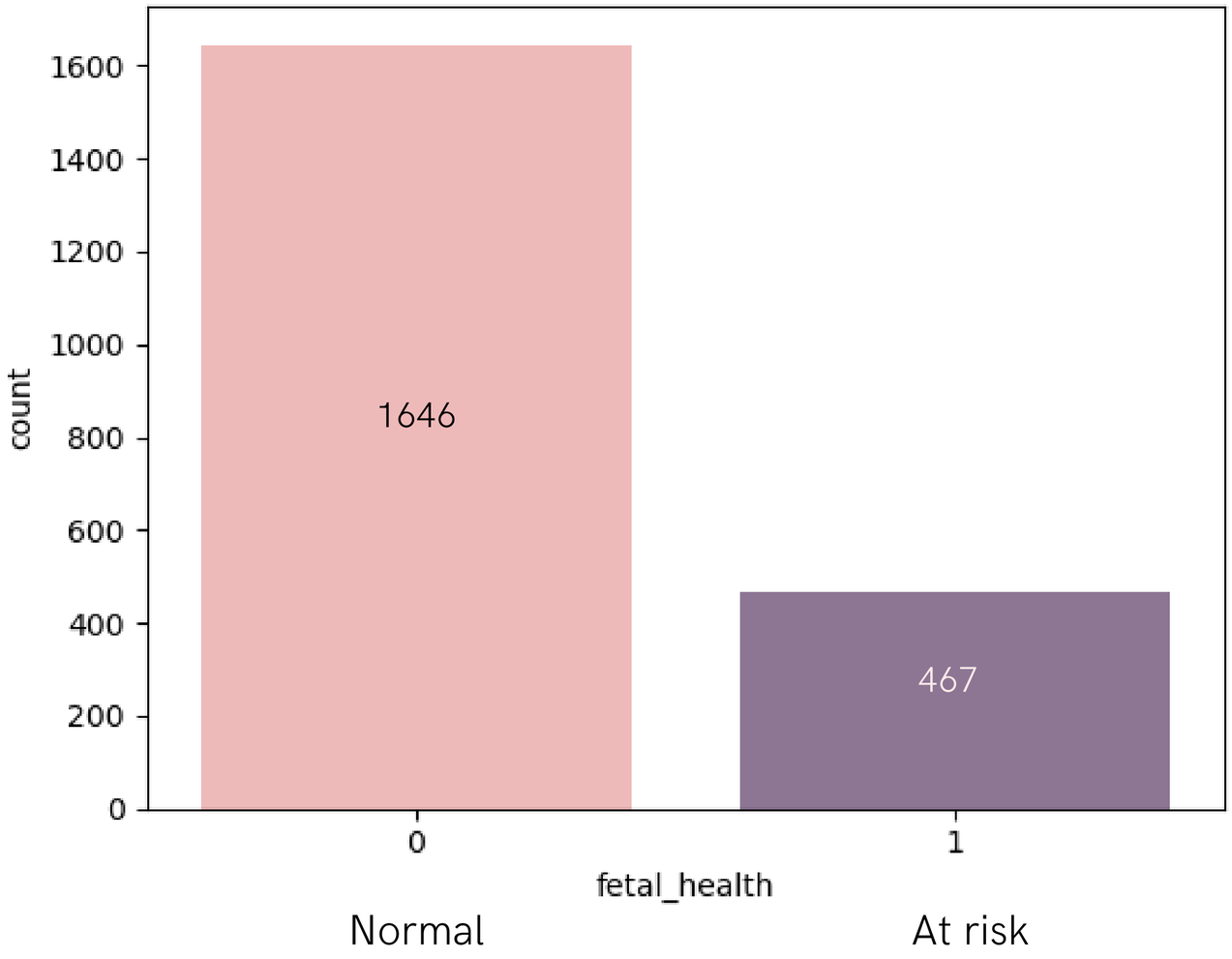
- The majority of the observed fetal heart rate acceleration, fetal_movement, and column decelerations were 0.00.
- The majority of fetal heart rates are 133 BPM
- abnormal_short_term_variability is a measure of the percentage of time that abnormal variability is observed. Most fetuses are said to be healthy but there are some records with a high percentage of abnormal short-term variability.

KDE Plot



- the baseline_value and histogram_median columns are symmetrical, besides that they tend to be positively skeww
- fetal_movement, light_decelerations, severe_decelerations, prolonged_decelerations, percentage_of_time_with_abnormal_long_term_variability, histogram_number_of_zeroes, positive skew extreme histogram_variance

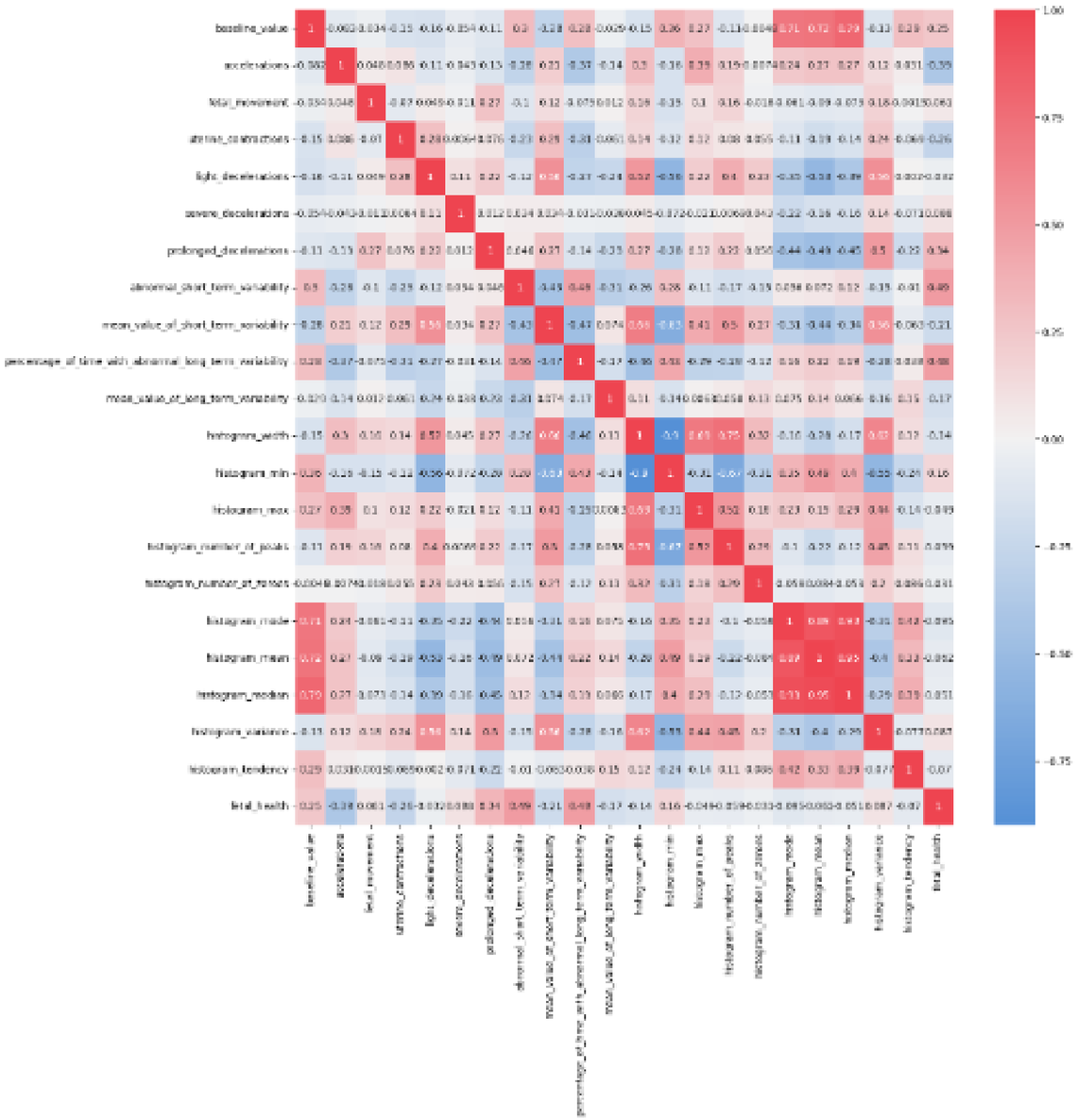
Multivariate Analysis



- The target column is changed to binary, by combining suspect and pathalogic

- A value of 0 means the fetus is normal and a value of 1 means the fetus is at risk

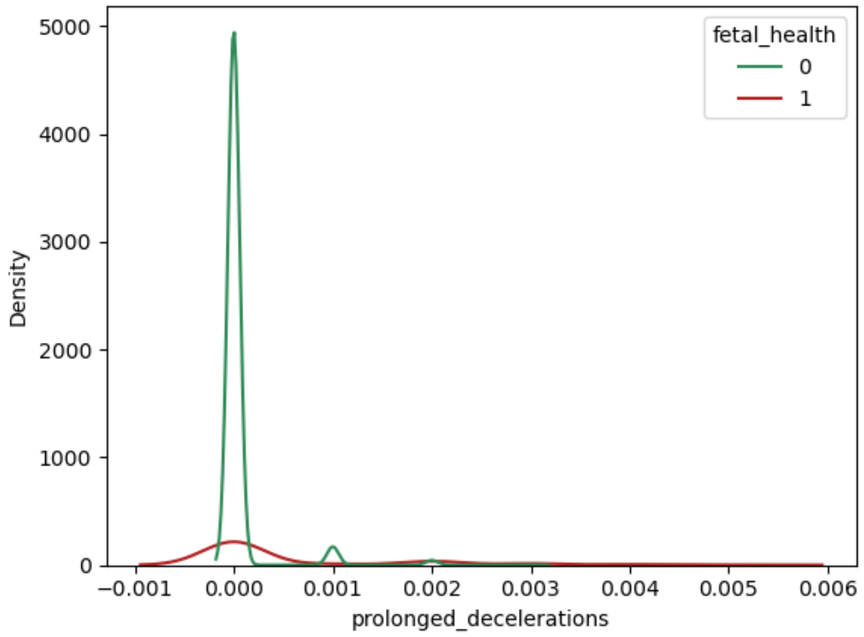
Heatmap correlation



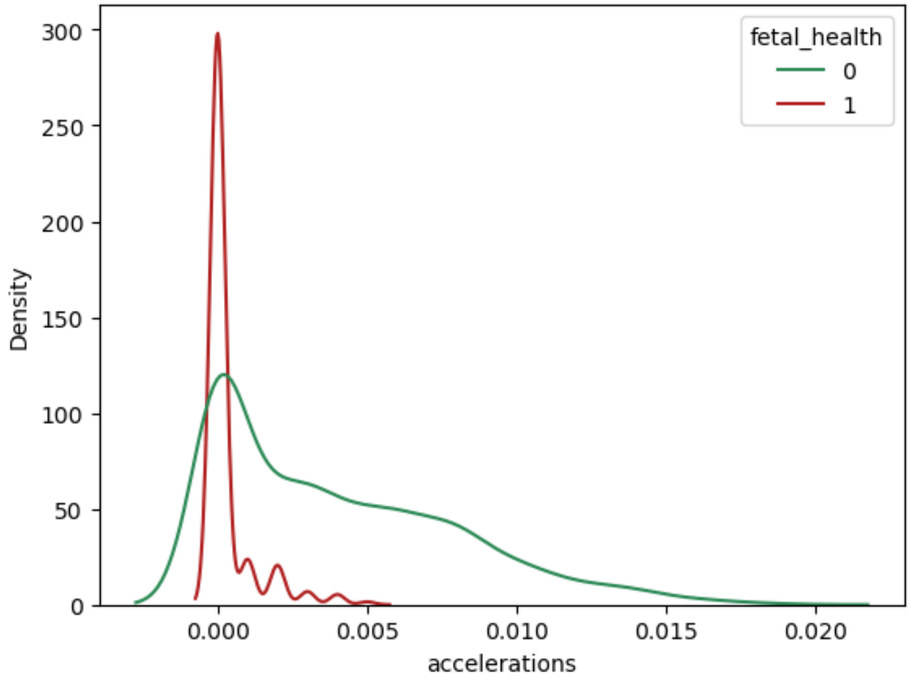
- Based on the correlation matrix, it is clear that `accelerations`, `prolonged_decelerations`, `abnormal_short_term_variability` and `percentage_of_time_with_abnormal_long_term_variability` are features with a high correlation with `fetal_health`.

- `histogram_mode`, `histogram_mean`, `histogram_median` are strongly correlated, this indicates that there is redundant information in the three columns

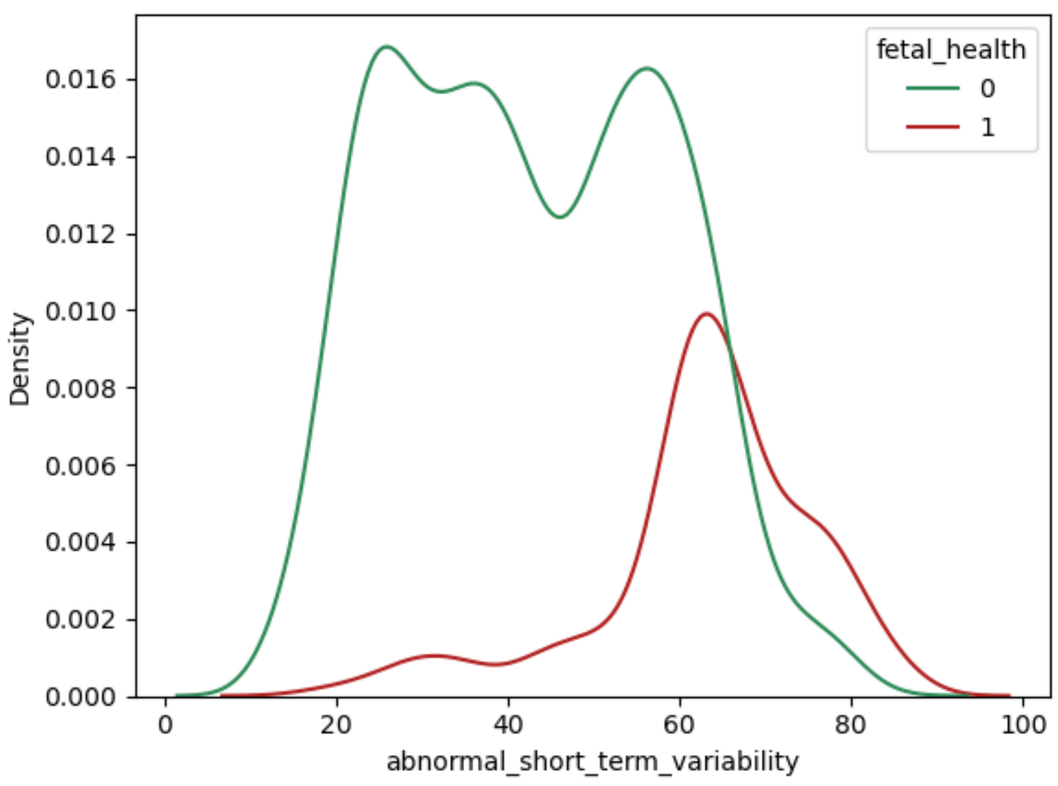
Accelerations & Prolonged_decelerations

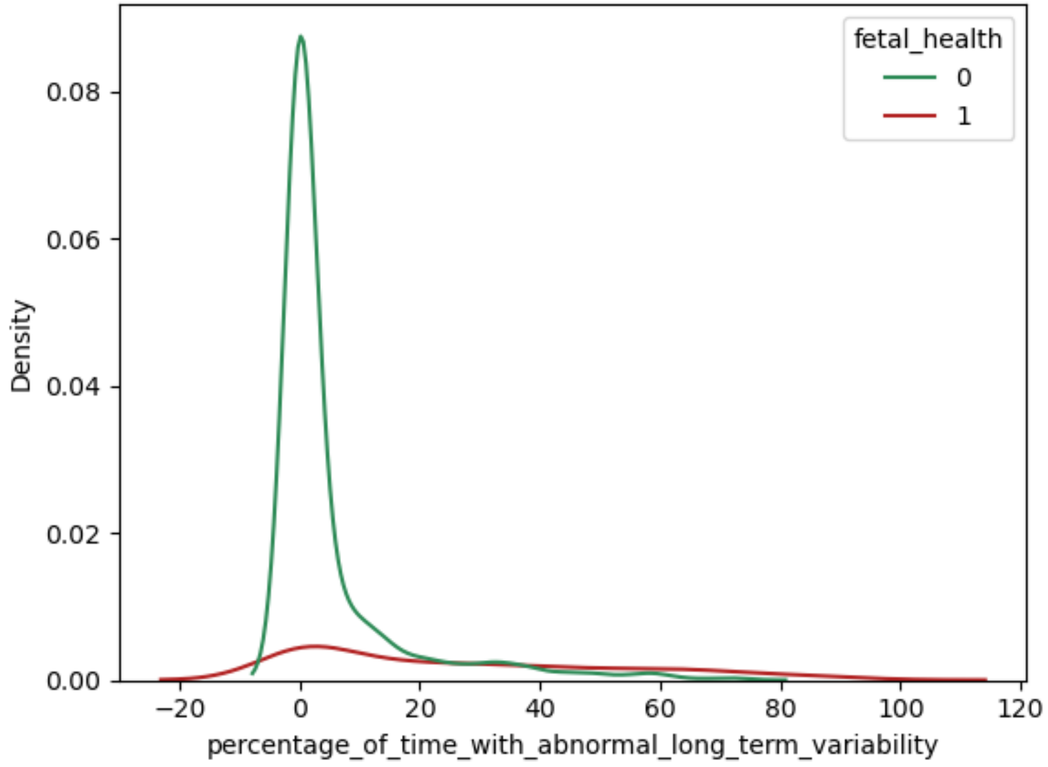


fetal_health	prolonged_decelerations	amount
0	0.000	1577
	0.001	54
	0.002	14
	0.003	1
1	0.000	358
	0.001	16
	0.002	58
	0.003	23
	0.004	9
	0.005	3



- A healthy fetus has a higher value of fetal accelerations per second than a fetus that is in danger
- prolonged_decelerations Fetuses at risk tend to be higher than healthy fetuses



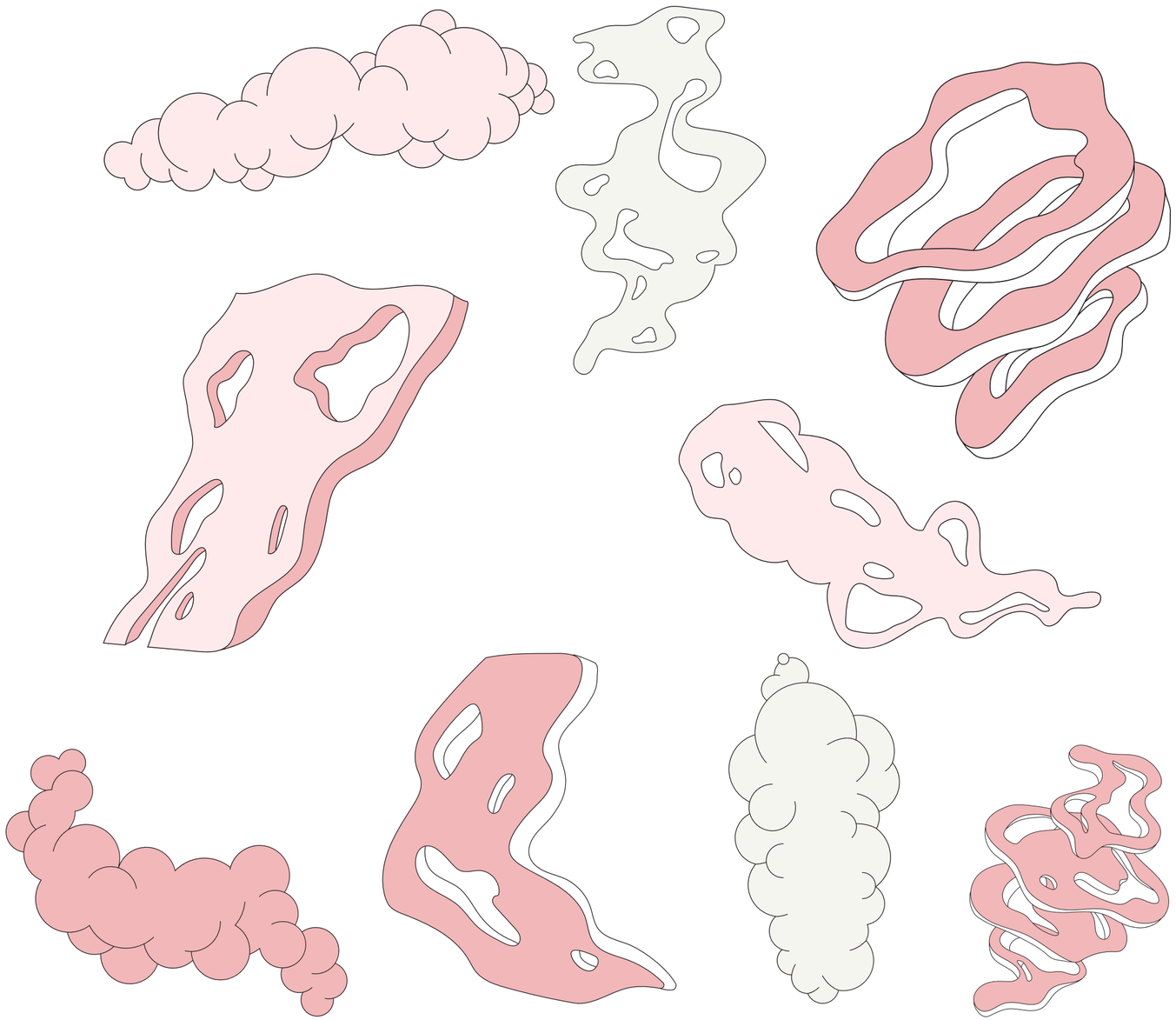


- according to what was mentioned above, the abnormal_short_term_variability percentage value for normal fetuses will be lower than fetuses at risk
- Abnormal_short_term_variability percentage values of normal fetuses are spread the most at +/- 23% while at risk fetuses at +/- 65%
- the percentage of abnormal long-term variability in both normal and at-risk fetuses is spread at most at 0%, but at-risk fetuses have a higher percentage of abnormal long-term variability

3

Modeling

Supervised machine learning classification



3 Models

- Logistic regression
- K-Neareast neighbors
- Random Forest

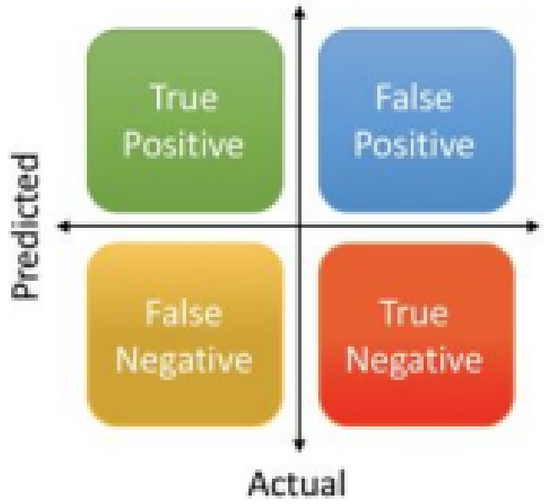
In classification task, we use the metrics of:

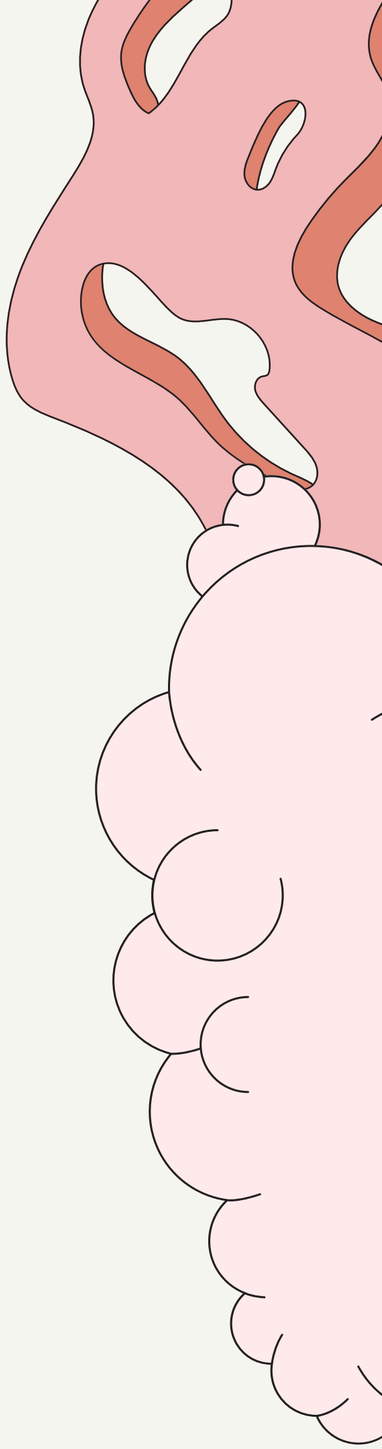
- Precision
- Recall
- F1-Score
- Accuracy

$$\textbf{Precision} = \frac{\text{True Positive}}{\text{Actual Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Positive}}$$

$$\textbf{Recall} = \frac{\text{True Positive}}{\text{Predicted Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Negative}}$$

$$\textbf{Accuracy} = \frac{\text{True Positive + True Negative}}{\text{Total}}$$





Target

0: Normal 1: At risk

In the case of fetal health, let's identify two types
FALSE :

- False Positive: we think that the fetus is at risk, but it is actually normal
- False Negative: we think that is normal, but it's actually a risk

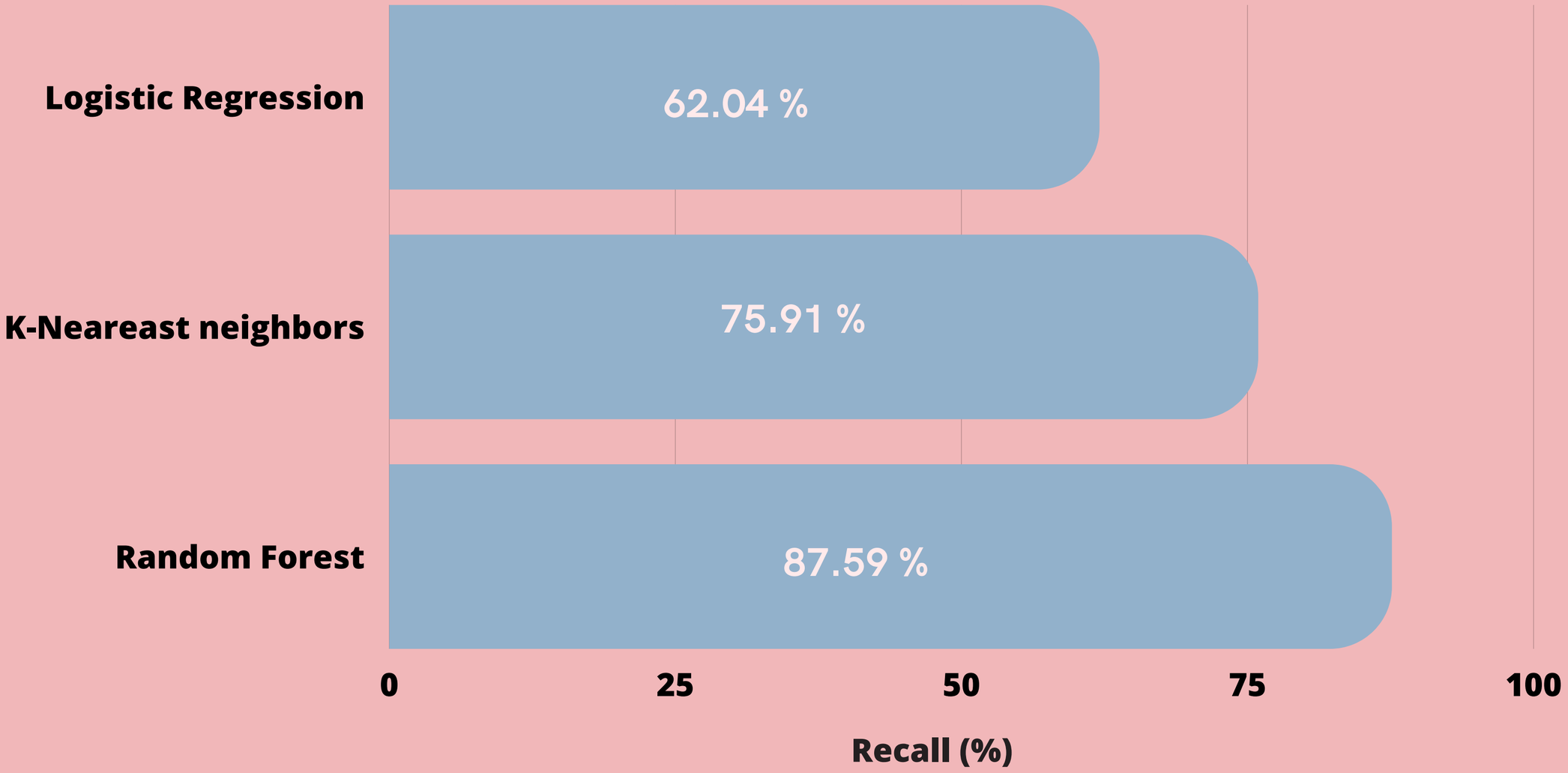
Primary Matrix Evaluation

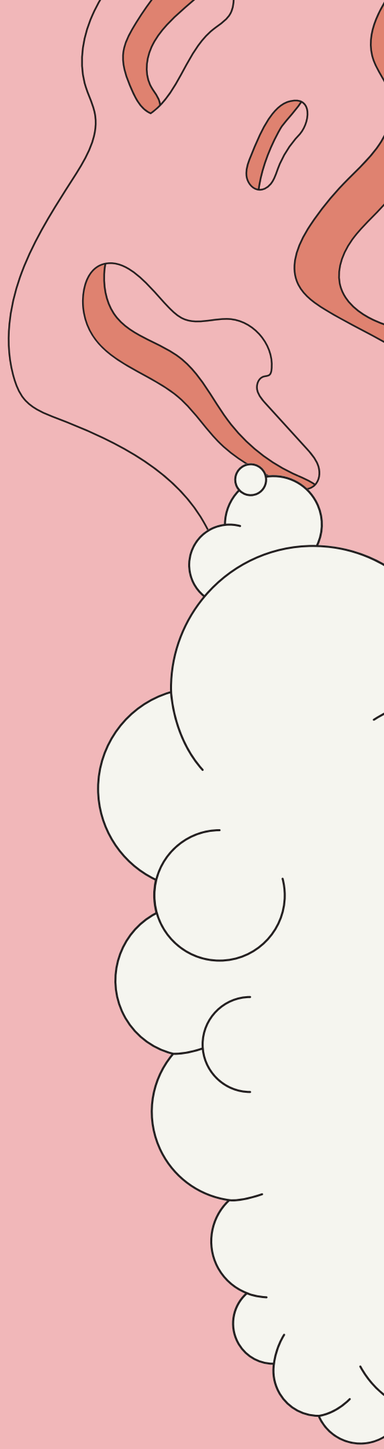
False Negative is a more ‘dangerous’ issue, and we would like our model to have as little false negatives as possible.

If we think fetus is doing normal, but it is risk, and we’re not prepared - that would be disastrous.

Therefore, the metric to highlight is recall, as it is the one that is connected to False Negative.

Baseline Model





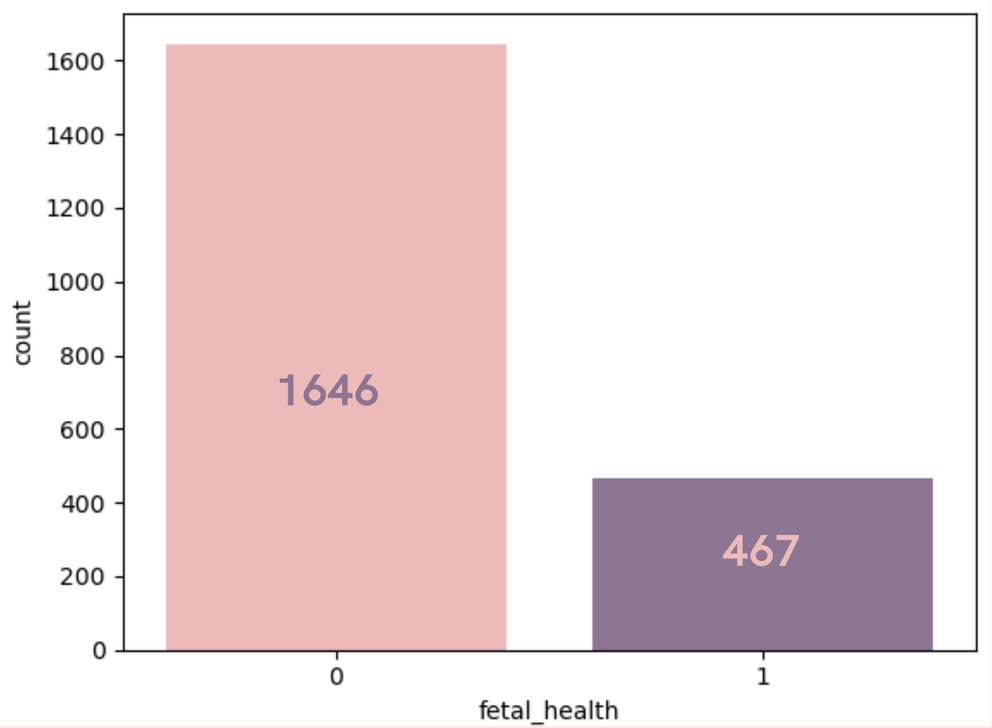
1 Multicolinearity Handling

- The columns histogram_mode, histogram_mean, histogram_median are highly correlated
- Keep the histogram_mode column because the correlation with the target is the highest

2 Scaling

- Because our data ranges are not the same, scaling is necessary
- The hope is faster computing and increased performance
- Using StandardScaler

3 Imbalance Handling



- The target column is still imbalanced
- Balancing targets with the SMOTE method - because the data is small and avoids duplicates

4 Hyperparameter tuning

Logistic regression

Penalty: ['l1', 'l2'],
C: np.logspace(-4,4,20)

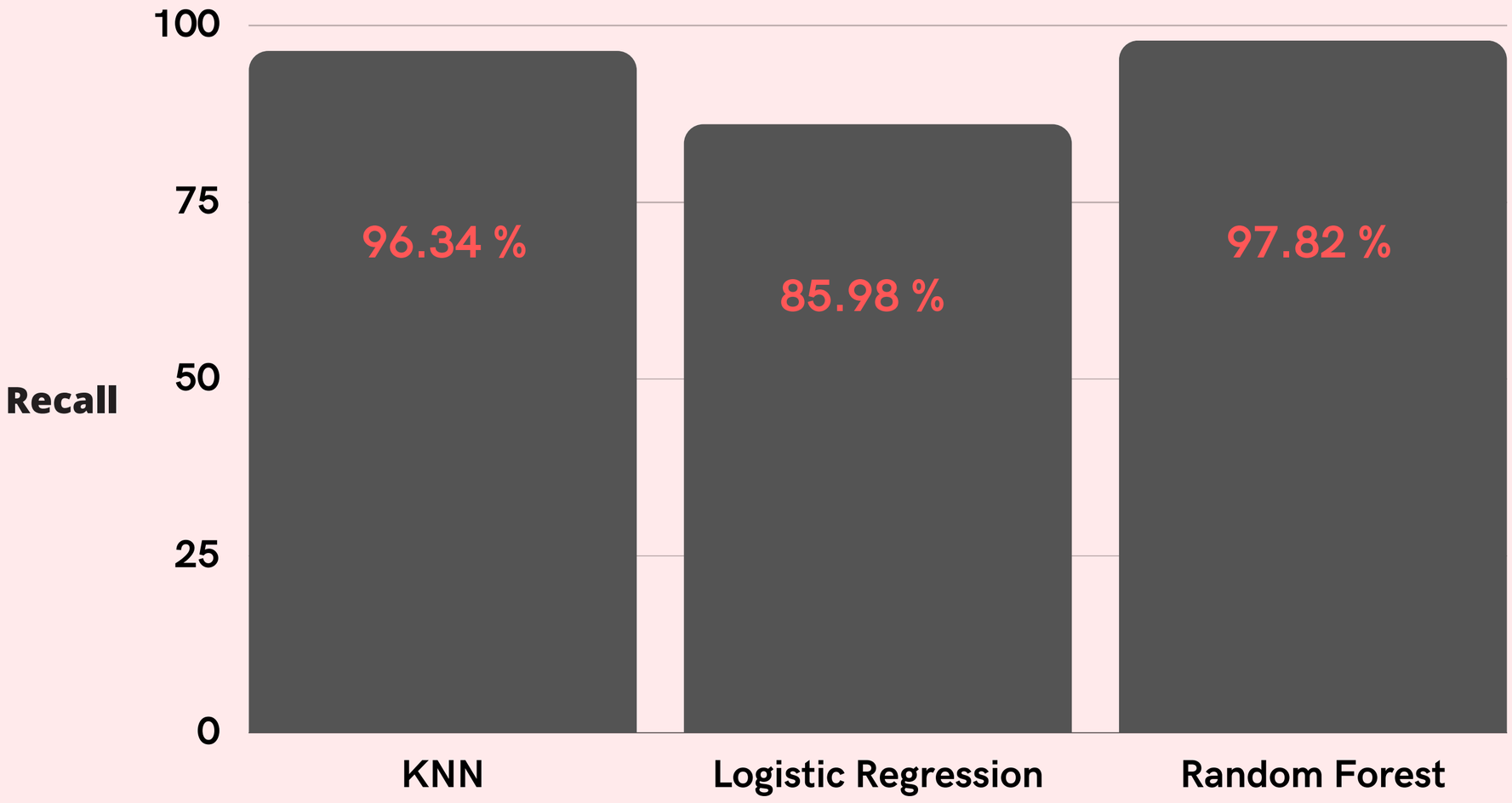
Random Forest

n_estimators: (10,20,30,40,50),
max_depth: (4,6,8,12,14,16)

K-Neareast neighbors

n_neighbors: (2,3,4,5,6,7,8)

Optimum model after Hyperparameter Tuning



KNN	Logistic Regression	Random Forest
n_neighbors': 3	{C': 10000.0, 'penalty': 'l2'	'max_depth': 16, 'n_estimators': 50

The Best Model

Random Forest

'max_depth': 16, 'n_estimators': 50

Evaluation

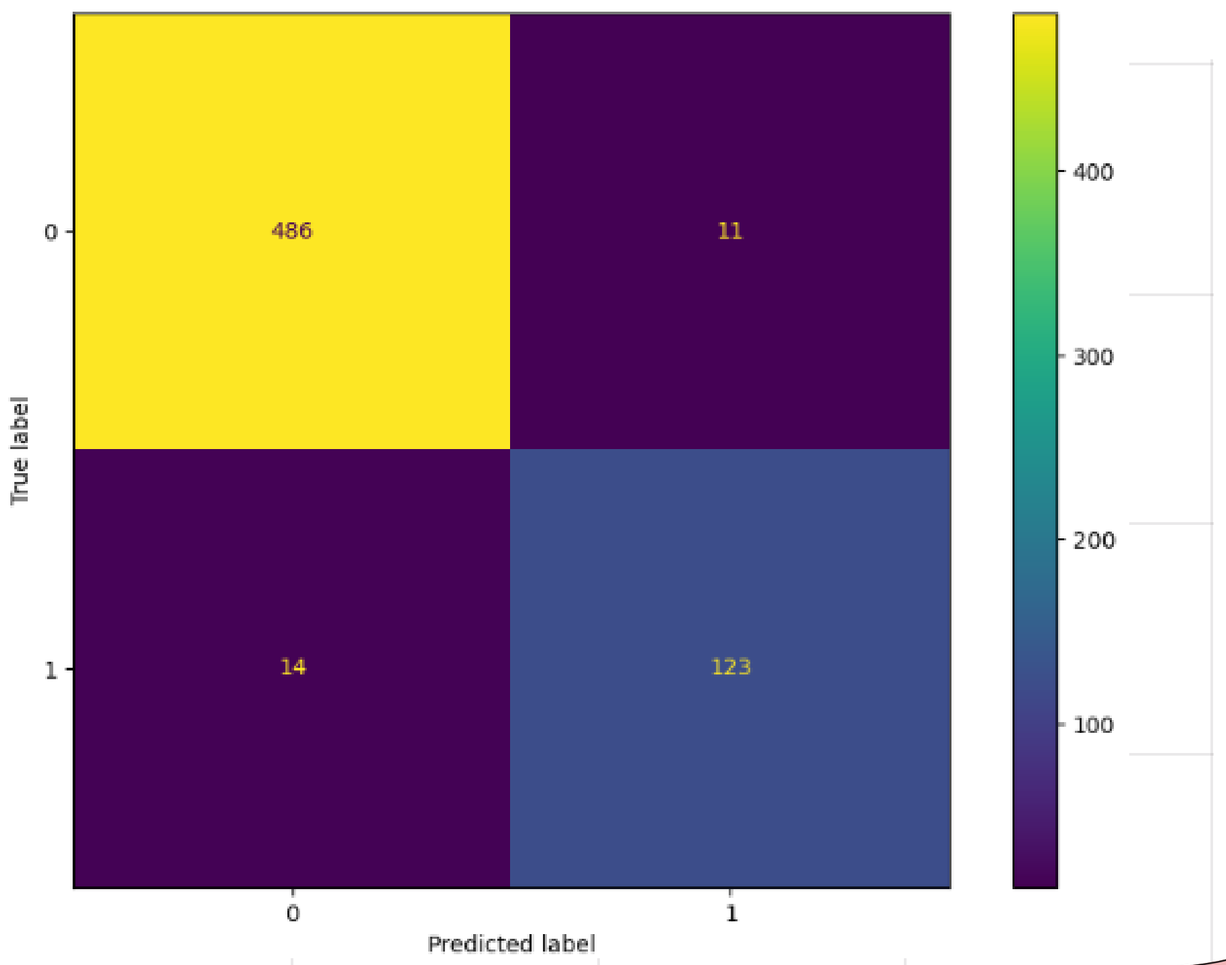
Using test data

0 : Normal 1 : At risk

- 486 True Negative
- 11 False Positive
- 14 False Negative
- 123 True Positive

Recall 90%

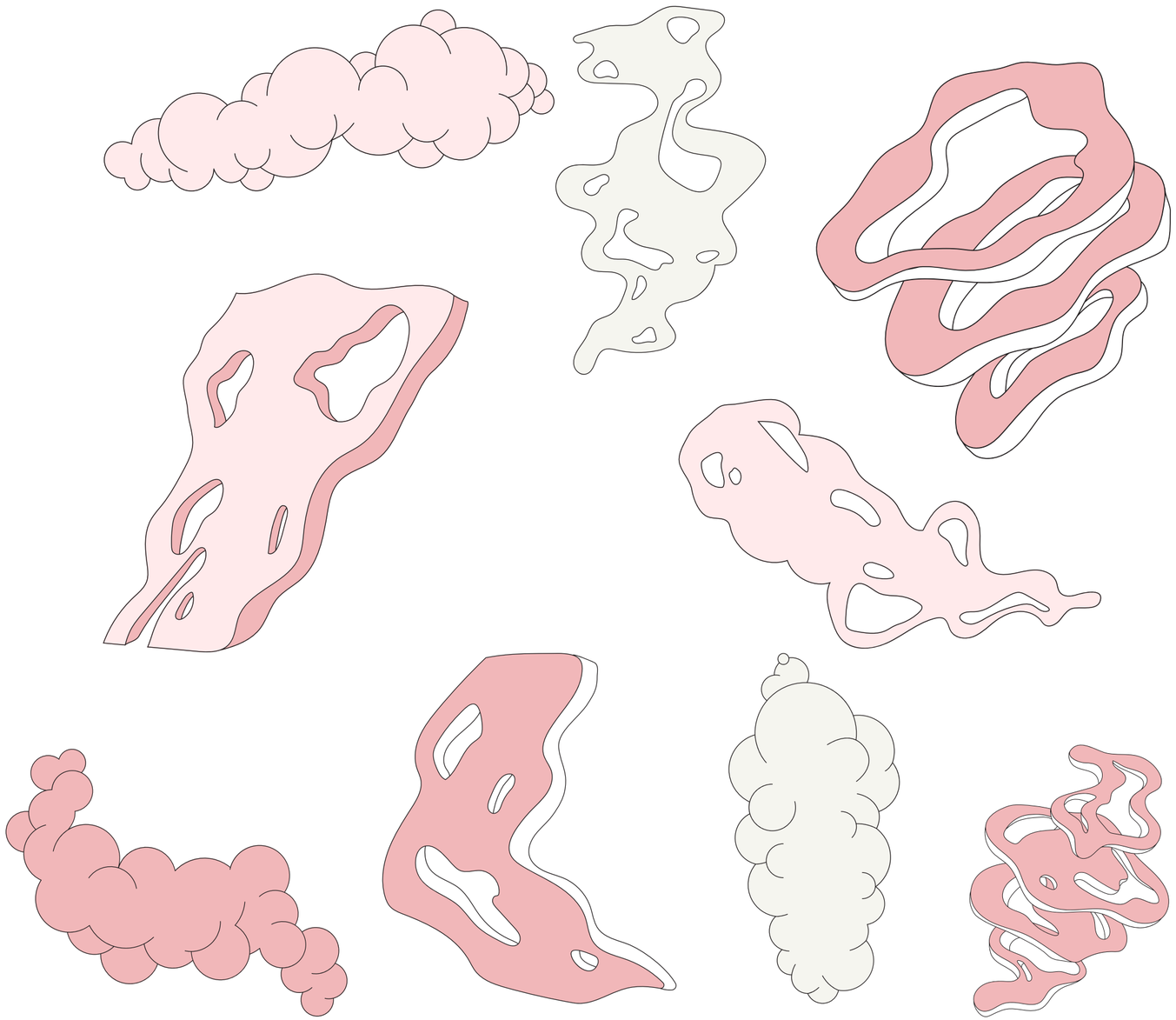
Accuracy 96%



4

Recomendation

Conclusion and
Recomendation

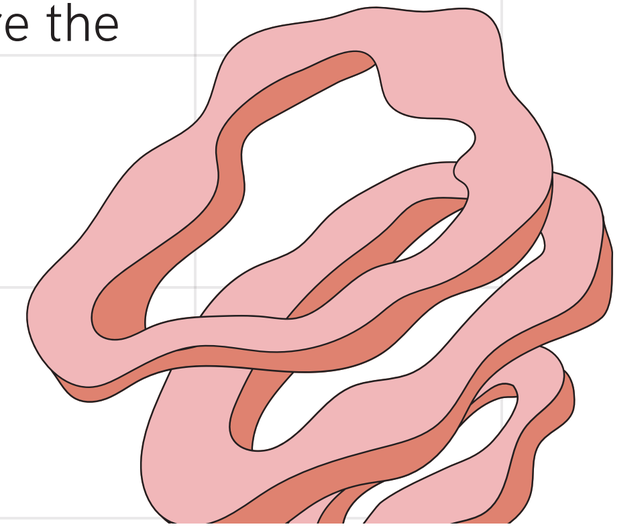


Conclusion

Cardiotocogram data provide easily accessible and interpretable insight into fetal health outcomes. The machine learning model is able to quickly predict if the fetus is in a distressed state using CTG data with high recall, precision and accuracy values. With the ability of machine learning models to determine the health of the fetus and the knowledge of obstetricians, it is hoped that the health of the fetus and mother can be known precisely so as to effectively reduce the problem of fetal death.

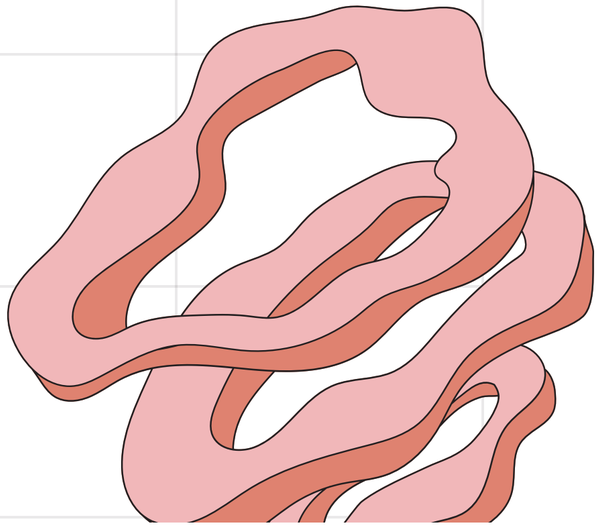
Recomendation

- Machine learning with CTG data is able to predict the health of the fetus and has a fairly good performance with a recall value of 90% and 96% accuracy. This can quickly detect the health condition of the fetus. I recommend that Machine learning be applied to the classification of fetal health, so that the analysis of CTG results is faster. Therefore, CTG can be done as often as possible in pregnant women because interpretation is faster, so that pregnant women and doctors know for sure the condition of the fetus so they can provide proper treatment



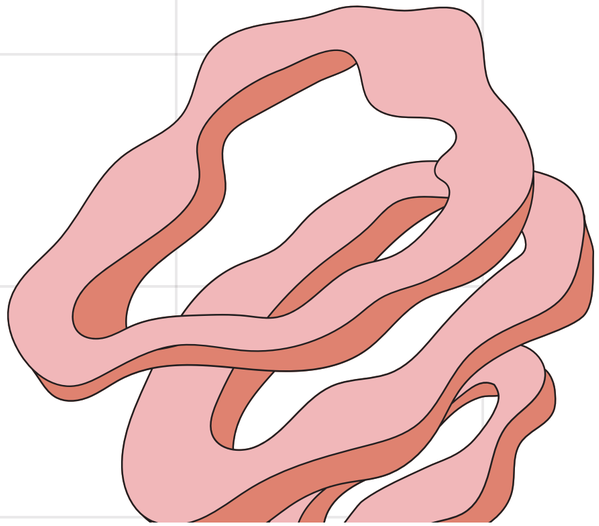
Recomendation

- I recommend that healthcare providers pay attention to all measures of fetal heart rate, as it is the strongest predictor of fetal health outcomes at risk. These measures include the percentage of time with abnormal short-term variability, the percentage of time with abnormal long-term variability, and prolonged decelerations.



Recomendation

- My last recommendation is the prediction of machine learning as a tool that makes it easier for obstetricians to determine fetal health based on CTG data, therefore it is necessary to double check these predictions because there are still deficiencies in this machine learning



Thank You

Reach Me

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